



“At Isro, gender has never dictated who gets the top assignments”

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sion,” says Ranjana.

122 | That said, there have been considerable improvements in policies like flexible work arrangements, parental leave, and wellness programmes. These initiatives are reshaping the workplace, making it more supportive for everyone. “Ultimately, it’s about creating an environment where people—whether men or women—can thrive, grow, and lead,” says Ranjana. “Look at sports—our women athletes are winning more medals than their male counterparts. Similarly, women have an excellent aptitude for science and technology.”

In DRDO, women are directors-general, directors and programme directors, spearheading R&D for defence tech. The government’s Department of Science & Technology runs many schemes that support women in STEM. Ranjana’s ambit includes 15 centres of excellence across institutions such as IITs and the Indian Institute of Science (IISc).

“Space isn’t just about launching rockets; it’s about launching dreams for future generations,” says Ranjana, who leads interdisciplinary projects involving satellite develop-

ment, ground stations, and market deployment of commercial offerings from space. Women scientists account for 30% of her teams, collaborating across institutions.

Women in India’s space programme are not restricted to technical roles; they also shape space law, policy frameworks, and entrepreneurship. “Women entrepreneurs are making strides in India’s space tech ecosystem, addressing challenges like cost-effective satellite manufacturing and space debris management,” Ranjana says.

SPACE FOR WOMEN

The journey is not without challenges. A nationwide government survey revealed that women constitute only 13% of scientists and science faculty at Indian higher education and research institutions. According to World Bank data, nearly 43% of STEM graduates in India are women, one of the highest shares globally, but women account for only 14% of scientists, engineers, and technologists at research institutes and universities.

This could be because many women engineers ease into management

roles. To Isro’s credit, it provides an environment that is challenging and collaborative enough to attract women into pure engineering roles. None of the women *BT* spoke to for this report were from IITs. Isro recruits from diverse technical institutes across India, giving students with varied backgrounds a chance.

Neelavathy says the data can improve only if the STEM drive gets more women in rural India. In international conferences hosted by the National Aeronautics & Space Administration, European Space Agency and Japan Aerospace Exploration Agency, she says the ratio of women to men from India is low. “While cities offer opportunities, women in rural areas still face hurdles. Many girls aren’t encouraged to pursue education beyond school,” she says.

“Government programmes like the Smart India Hackathon mandate at least one female participant per team, which helps boost inclusivity,” Neelavathy says. Isro also engages with young minds to inspire interest in space exploration. Programmes like “Yuvika” provide schoolchildren with hands-on learning about space



ON A MISSION

Some of the key women scientists currently working on India’s upcoming space missions